

Xinyue (Cindy) Wang

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Bachelors in Computer Science and Information Systems

May 2028

- **Relevant Coursework:** *Application + Database Design and Development, Computer Systems, Software Engineering, Artificial Intelligence, Functional Programming, Imperative Computation*
- **Activities/Interests:** *CMU Club Volleyball, GroupX Dance, Chinese Student Association, Baking*

EXPERIENCE

Software Engineer Intern

Jun 2026 – Present

LeapBound AI

San Jose, CA

- Independently architected and shipped **2 production full stack AI applications** at leapbound.ai for analyzing search visibility and AI brand visibility, delivering client reports with prioritized recommendations.
- Engineered scalable infrastructure using **Python, FastAPI, Next.js, Supabase (PostgreSQL), and Cloudflare**, with asynchronous services, LLM-assisted processing pipelines, and integrations with **5 external APIs**.
- Optimized the production crawl pipeline for websites with up to **1,000 discovered URLs**, improving crawl performance by **3x** through a redesigned page selection strategy.

Software Engineer Intern

Apr 2026 – Jun 2026

Howbe Technology

San Jose, CA

- Engineered production features for a **cross-platform Flutter application** (App: WhoBe) with a **FastAPI backend**, delivering backend services and cloud-integrated application workflows.
- Streamlined backend performance by designing **REST APIs on Google Cloud Run**, replacing **O(n) client requests** with a single bulk aggregation endpoint to minimize network overhead.
- Resolved caching and state synchronization race conditions, including a silent data loss bug, improving reliability with **30+ unit and integration tests** across **Flutter** and **pytest**.

Undergraduate Researcher – Cyber Defense Automation

Sept 2024 – Present

CMU SCS, Department of Defense, under Profs. David Garland and Ehab Al-Shaer

Pittsburgh, PA

- Researched **automated cyber defense playbook generation** and knowledge graph construction across **183 MITRE D3FEND** techniques; currently co-authoring an **IEEE conference paper**.
- Engineered a **RAG framework** leveraging **LLMs** and agentic reasoning to parse action semantics and mitigate hallucinations.
- Developed **Python tooling** for **cross-LLM validation** and automated quality assessment, achieving **88.1% precondition** and **89.8% postcondition extraction accuracy**.

PROJECTS

Creamery Management Platform | *Ruby on Rails, React, JavaScript, SQL, REST APIs, Minitest, Cucumber*

- Built a full-stack multi-store retail chain management platform through **test-driven development**, achieving **100% code coverage** with **250+ automated unit tests and behavior-driven Cucumber tests**.
- Developed the application using **Ruby on Rails (MVC)** and **React**, designing a relational database schema with **8 models**, implementing a versioned **JSON REST API** with **15+ endpoints**, and building interactive dashboards with a **live time-clock** for payroll calculation.
- Configured **session-based authentication** with **bcrypt** and **role-based authorization** using **CanCanCan**, securing **13 controllers** and **50+ views** across **3 permission tiers**.

Memory Allocator | *C, Segregated Free Lists, GDB, Data Structures, Memory Optimization*

- Programmed a **memory management system** in C implementing malloc, free, realloc, and calloc using **segregated free lists**.
- Optimized memory efficiency by minimizing data layout overhead, achieving **75%+ heap utilization** and **10,000+ KOPS throughput**.
- Implemented runtime consistency checks to validate memory safety, allocator correctness, and data structure integrity across a **64-bit address space**.

TECHNICAL SKILLS

Languages: Python, C, SQL (PostgreSQL), Java, JavaScript, HTML/CSS, Ruby, Assembly, R

Tools: React, Next.js, FastAPI, Flutter, Ruby on Rails, Git, Docker, CI/CD, Supabase, Cloudflare, REST APIs